EXPLORATORY DATA ANALYSIS (EDA) & DATA CLEANING ON TITANIC DATASET BY VIDHYA P

1. Introduction

Exploratory Data Analysis (EDA) is an important step in the Data Science lifecycle. It helps understand the structure, quality, and patterns present in a dataset before performing advanced analysis or machine learning. Data cleaning is equally important because real-world datasets often contain missing values, duplicate records, and inconsistencies.

The objective of this project is to perform data cleaning and exploratory data analysis on a Titanic dataset using Python, Pandas, Matplotlib, and Seaborn. The project focuses on identifying missing values, handling inconsistencies, detecting outliers, and visualizing important patterns within the data.

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2. Dataset Overview

Dataset Name

Titanic Dataset (Sample Dataset)

Dataset Source

A sample Titanic passenger dataset in CSV format was used for this project.

Dataset Size

- Number of Rows: 20
- Number of Columns: 7

Dataset Columns

1. PassengerId
2. Survived
3. Pclass
4. Sex
5. Age
6. Fare
7. Embarked

Project Objective

The objective of this project is to clean the dataset, identify missing values and outliers, perform exploratory data analysis, and derive meaningful insights from the data.

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3. Dataset Inspection

The dataset was loaded into a Jupyter Notebook using the Pandas library.

The following functions were used to inspect the dataset:

- head()
- shape
- info()
- describe()

These functions helped understand:

- Dataset dimensions
- Column names
- Data types
- Statistical summaries
- Missing values

The dataset contains both numerical and categorical features.

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4. Data Profiling and Missing Value Treatment

Missing Value Identification

Missing values were identified using:

- isnull()
- sum()

The Age column contained missing values.

Missing Value Percentage Calculation

The percentage of missing values was calculated using:

Missing Value Percentage = (Missing Values / Total Rows) × 100

This helped understand the quality of data before cleaning.

Missing Value Handling

The missing values in the Age column were replaced using the median value of the column.

Median imputation was selected because it is less affected by outliers and preserves the overall distribution of the data.

For categorical columns, mode-based imputation can be applied when required.

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5. Duplicate Detection and Removal

Duplicate records can affect the quality and accuracy of analysis.

The duplicated() function was used to identify duplicate rows.

Any duplicate records found were removed using the drop_duplicates() function.

This ensured that only unique observations remained in the dataset.

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6. Outlier Detection and Visualization

Outliers are observations that differ significantly from other data points.

A box plot was created for the Fare column to identify extreme values.

Outlier Detection Method

The Interquartile Range (IQR) method was used.

Steps followed:

1. Calculate Q1 (25th percentile)
2. Calculate Q3 (75th percentile)
3. Compute IQR = Q3 − Q1
4. Calculate lower and upper limits
5. Remove values outside the acceptable range

This process improved the reliability of the dataset.

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7. Exploratory Data Analysis (EDA)

Several visualizations were created using Matplotlib and Seaborn.

Survival Distribution

A count plot was generated to analyze the number of passengers who survived and those who did not survive.

Gender Distribution

A count plot was used to visualize the distribution of male and female passengers.

Age Distribution

A histogram was created to understand the age distribution of passengers.

Correlation Matrix Heatmap

A heatmap was generated to identify relationships among numerical variables in the dataset.

The heatmap provided a visual representation of correlations between features.

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8. Correlation Analysis

The correlation matrix helped identify relationships between numerical variables.

Top Three Observations

1. Passenger class showed a relationship with survival status.
2. Fare and passenger class displayed a noticeable relationship, indicating that higher-class passengers generally paid higher fares.
3. Age showed a relatively weak relationship with survival compared to passenger class and fare.

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9. Key Insights

The following insights were obtained from the analysis:

1. Female passengers had a higher survival rate than male passengers.
2. Most passengers were between 20 and 40 years of age.
3. Passenger class influenced survival outcomes.
4. Missing values were successfully handled using median imputation.
5. Outlier detection improved data quality.
6. Data cleaning made the dataset more reliable for analysis.

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10. Challenges Faced

During this project, several challenges were encountered:

- Understanding missing value treatment techniques.
- Calculating missing value percentages.
- Detecting outliers using statistical methods.
- Applying the IQR method correctly.
- Creating visualizations using Matplotlib and Seaborn.
- Interpreting correlation patterns from the heatmap.

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11. Conclusion

This project successfully demonstrated Exploratory Data Analysis (EDA) and Data Cleaning using Python.

The dataset was inspected, cleaned, and analyzed systematically. Missing values were handled using median imputation, duplicate records were removed, and outliers were detected using box plots and treated using the IQR method.

Several visualizations were created to understand patterns within the dataset. Important insights regarding survival status, age distribution, gender distribution, and passenger class were obtained.

This project improved practical knowledge of data preprocessing, data visualization, and exploratory data analysis, which are essential skills in Data Science.

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12. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

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13. References

- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Titanic Dataset